DriveCompose: Compositional Generative Reconstruction of Driving Scenes via Joint Completion

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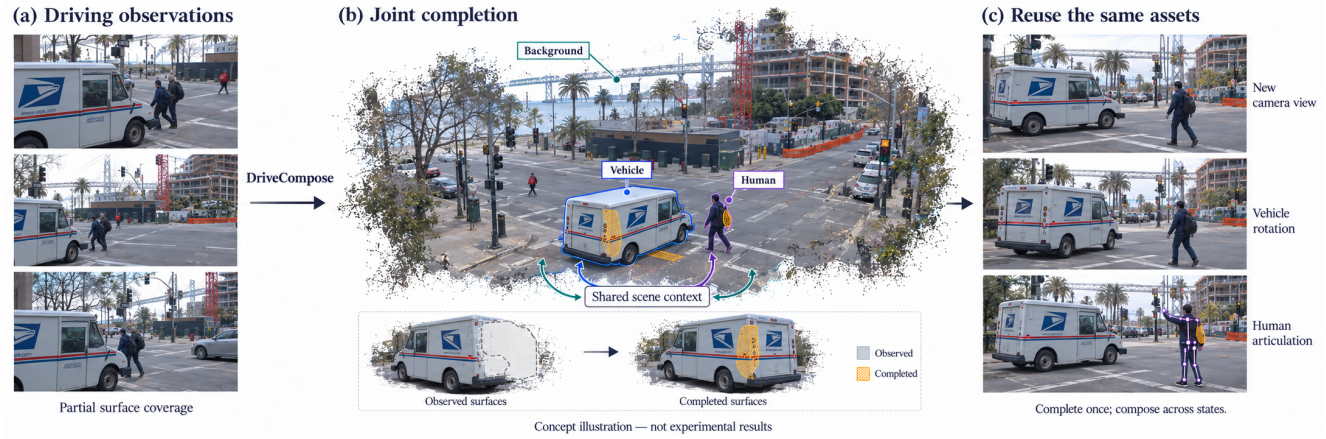

Figure 1. **DriveCompose concept.** DriveCompose jointly completes background, vehicle, and human content from driving images to form a compositional Gaussian scene. The same assets are reused under changes in camera viewpoint, vehicle pose, and human articulation. Gold marks completed regions.

Abstract

Driving simulation and scene editing require reconstructed scenes that remain usable when cameras and actors move beyond their recorded configurations. Camera or actor motion exposes previously unseen surfaces, requiring their geometry and appearance to be completed and assigned to the corresponding background, rigid asset, or human. We present DriveCompose, a framework for generative reconstruction of compositional driving scenes from image sequences. Visual features are aggregated in world coordinates for the background and canonical frames for rigid assets and humans, then encoded into component-specific latent representations. A conditional generative model trained with flow matching uses cross-component memory to share scene context and jointly complete the missing codes in these representations. Completion is constrained by the input observations to preserve the geometry and appearance of observed surfaces. On Waymo, DriveCompose improves dynamic scene reconstruction and off-trajectory view synthesis over the evaluated baselines. Type-specific decoders convert the completed latents into Gaussian components whose rigid and skeletal bindings support composition under different camera and actor states.

1. Introduction

Driving simulation and scene editing require 3D scenes that support changes in camera viewpoint, vehicle motion, and human pose [46, 49]. However, driving images reveal only the surfaces visible from the recorded viewpoints and actor configurations [1, 31]. Moving a camera, turning a vehicle, or changing a pedestrian's pose exposes previously unobserved regions (Fig. 2). Reconstructing scenes for these operations therefore requires completing the missing geometry and appearance while ensuring that completed vehicle and human surfaces move with their respective actors.

Recent feed-forward methods fuse multiple images to directly predict driving-scene geometry and Gaussian representations [3, 6, 45, 48]. Motion estimation and visibility modeling further help align dynamic objects and integrate observations over time. Multi-frame aggregation increases

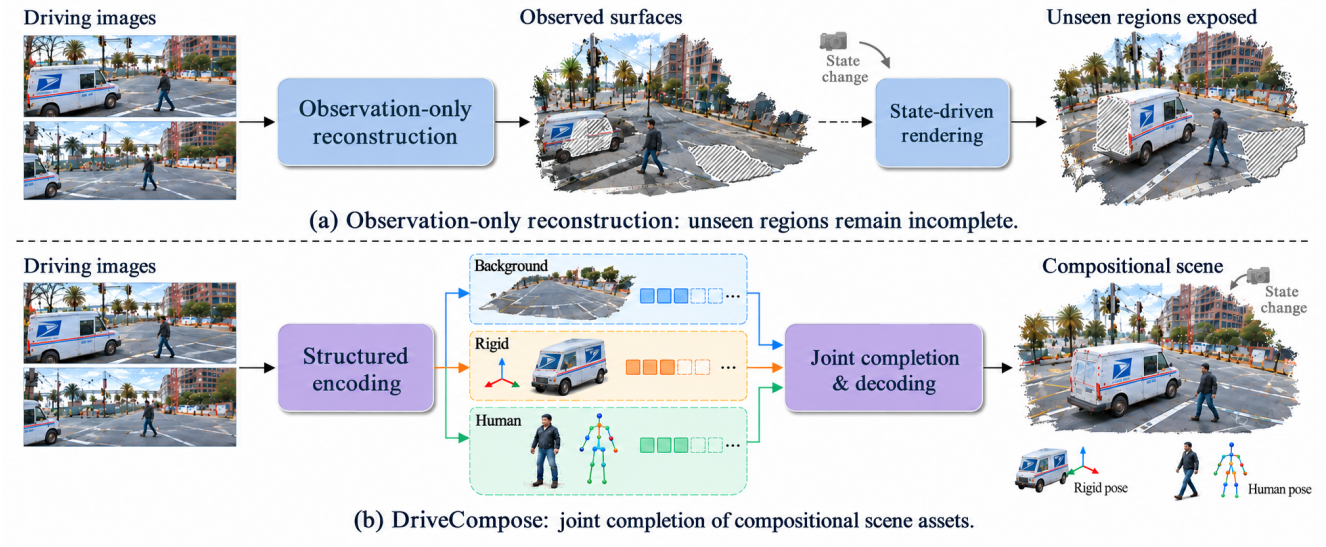

Figure 2. **From observed surfaces to compositional scene assets.** (a) Observation-only reconstruction leaves unseen surfaces incomplete when the viewpoint or actor state changes. (b) DriveCompose jointly completes the background and canonical rigid and human assets, then composes them under the requested state. Scene images are schematic illustrations.

coverage of surfaces visible in the input sequence. When a new viewpoint or actor motion reveals surfaces unseen throughout the sequence, reconstruction must infer their geometry and appearance.

Generative reconstruction uses asset priors, Gaussian completion, or generated views to infer unobserved content [17, 44, 47, 51]. Meanwhile, compositional reconstruction represents the background and moving actors separately to support vehicle motion and human pose control [5, 43, 49]. Applying generative completion to such scenes requires generated surfaces to move with their actors. When an actor is only partially observed, information from the background and other actors can provide additional context for inferring its missing content.

To this end, we introduce DriveCompose, which reconstructs driving image sequences into a generatively completed 3D scene with independently controllable actors. We represent the background in world coordinates, rigid surfaces in each asset's own coordinate frame, and human surfaces in a reference pose with skeletal bindings [5, 43]. Completion operates directly on these representations, allowing generated rigid and human surfaces to follow their actors' motion and pose changes (Sec. 3.1). The model jointly completes missing background and actor content using shared scene context.

Figure 3 follows the reconstruction pipeline. A visual encoder extracts image features and estimates scene geometry and actor motion. Motion-aware aggregation aligns features across views and time in the background and canonical asset frames, and type-specific encoders map them into a shared latent dimension. Local attention captures structure within each component. Cross-component memory gathers background and actor features and returns the shared scene context to each component, allowing missing surfaces to be inferred from both local observations and their surroundings. The model learns this joint completion through flow matching [18] while retaining observed latent codes.

Type-specific Gaussian decoders turn the completed codes into a background, canonical rigid assets, and articulated humans [15]. Rigid geometry follows asset poses, and human geometry follows skeletal motion through its skinning weights. Scene composition places these components in world coordinates for the requested camera and actor states, reusing the same completed content across state changes (Sec. 3.7).

On Waymo, DriveCompose improves dynamic scene reconstruction and off-trajectory view synthesis over the evaluated baselines (Sec. 4).

Our contributions are:

- We introduce DriveCompose, a generative reconstruction framework for closed-loop driving simulation. It recovers and completes backgrounds, vehicles, and humans from image sequences to construct compositional Gaussian scene assets. The resulting scene reuses the same completed assets for composition and rendering as camera poses, vehicle poses, and human poses change.
- We develop a cross-component latent completion model based on conditional flow matching. Local attention models spatial relationships within each component, while cross-component memory aggregates background and actor features to provide scene context for generating each component's missing content.

- We evaluate the framework and its key mechanisms through dynamic reconstruction, off-trajectory view synthesis, and component ablations. Experiments show that DriveCompose outperforms the evaluated baselines on reconstruction and view synthesis.

2. Related work

Driving-scene reconstruction. NeRF and Gaussian splatting provide the radiance representations underlying urban reconstruction [15, 22], with driving-specific rendering developed in NeuRAD and SplatAD [11, 33]. Feed-forward methods learn geometry and Gaussian prediction from images [2, 4, 12, 35, 36]. For driving logs, STORM aggregates dynamic Gaussians through scene flow, while DGGT combines camera recovery with temporal visibility and motion-guided interpolation [3, 45]. DrivingForward, StreetForward, PointForward, and Instant NuRec further explore surround-view inputs, temporal processing, point-aligned representations, and simulation-oriented reconstruction [6, 14, 32, 48]. TokenGS also decouples Gaussian prediction from pixels through learnable tokens [29]. DriveCompose aggregates recovered evidence onto background and canonical asset supports, then conditionally generates their missing content before Gaussian decoding.

Compositional scenes and controllable assets. Neural Scene Graphs, Street Gaussians, and DrivingGaussian separate scene content into background and moving components [23, 43, 50]. OmniRe includes canonical representations for rigid and articulated actors, while UniSim and HUGSIM connect reconstructed content to closed-loop simulation [5, 46, 49]. At the asset level, SMPL-based body models and animatable Gaussian methods provide canonical surfaces and motion bindings [13, 16, 20, 25, 27]; LHM and IDOL infer animatable humans, and UrbanCAD targets controllable vehicles [21, 28, 52]. These works establish the representation and control mechanisms used to manipulate scene components. Our completion variables retain this component structure, so generated surfaces inherit an asset identity, coordinate frame, and motion binding.

Generative reconstruction and completion. Image-conditioned 3D generation uses generated views [9, 19, 30] or structured latents encoding geometry and appearance [41, 42]. GenAssets learns driving-asset priors, CompleteSplat completes hidden Gaussian content, and JRM aggregates unaligned object observations through latent flow matching [17, 40, 47]. LatentWorld organizes semantic occupancy generation around entity and background latents [24]. DriveCompose targets RGB-renderable reconstruction conditioned on driving logs: a shared completion model exchanges context across world-space background and canonical rigid and human codes. Observed codes and known free space condition the completion, and type-specific decoders preserve each component's spatial frame and motion binding.

Generative driving scenes and videos. MagicDrive, DriveDreamer, and Vista generate driving imagery conditioned on scene structure or control [8, 10, 37]. StreetCrafter uses generated views to supervise dynamic Gaussian reconstruction; HorizonForge combines scene editing with video diffusion [38, 44]. WorldSplat generates Gaussians and refines their rendered videos, while DIFIX3D+ applies diffusion-based image refinement [39, 51]. These methods connect generative imagery and three-dimensional scene representations at different stages. DriveCompose performs completion during scene construction and stores the resulting background and canonical assets; subsequent camera, rigid-body, and skeletal states render the same completed content.

3. Method

DriveCompose converts multi-view image sequences into a compositional Gaussian scene through visual encoding, canonical feature aggregation, latent encoding, joint latent completion, type-specific Gaussian decoding, and scene composition (Fig. 3). Canonical aggregation aligns observations of each component across time; joint completion uses the surrounding scene to infer its missing geometry and appearance.

3.1. Problem formulation and compositional representation

The input is a synchronized image log $\mathbf{I}_{t,v}$ with timestamps, camera intrinsics, and camera-to-rig calibration. Rig poses are provided or estimated relative to the first frame. Coordinate conventions and image preprocessing are detailed in Appendix D of the supplementary material.

The output is a compositional Gaussian scene

$$\mathcal{W} = \big(\mathcal{G}_b, C_{\text{sky}}, V_{\text{sky}}, \{\mathcal{A}_a\}, \mathcal{B}, \xi\big), \tag{1}$$

where $\mathcal{G}_b$ is the world-space background, C_{sky} a sky cubemap, V_{sky} its supported directions, $\mathcal{A}_a$ the canonical asset for identity a, $\mathcal{B}$ the finite spatial support, and ξ the sampling seed. The components are rendered by Gaussian splatting [15]. Vehicles are represented as rigid assets with fixed local geometry and dimensions. Humans additionally retain a rest skeleton and skinning weights. Scene state controls the placement and articulation of these fixed components.

We model missing content through conditional generation. Let $\mathcal{Q}$ be the recovered query layout and C its observed evidence. The completion model samples unknown

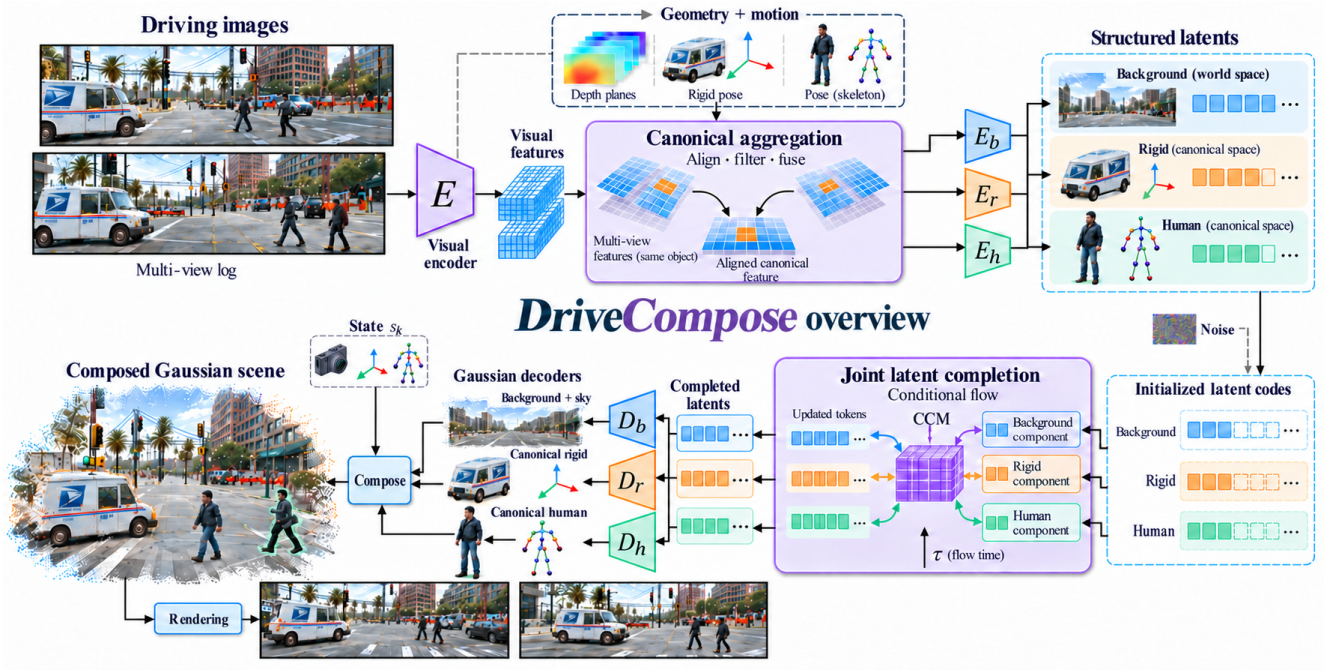

Figure 3. **DriveCompose overview.** Visual encoding and motion-aware canonical aggregation recover component features, which type-specific encoders map to structured latents. Conditional-flow completion infers unknown content using cross-component context. Gaussian decoders produce background, rigid and human components for state-driven composition and rendering. Scene and asset images are schematic illustrations.

codes $\mathbf{z}_u$, while the codec D_ψ decodes the assembled codes into scene content:

$$\mathbf{z}_u \sim p_\theta(\mathbf{z}_u \mid C, \mathcal{Q}), \qquad \mathcal{W} = D_\psi(\mathbf{z}^{\text{obs}}, \mathbf{z}_u; \mathcal{Q}). \tag{2}$$

The query layout assigns each latent code to a spatial location and scene component. Completion fills the unknown codes within this layout, and decoding preserves their component identities and motion bindings.

3.2. Visual encoding

The Driving Log Encoder extracts patch features from each image and exchanges pooled context across cameras and timestamps [7, 34]. Geometry heads predict camera-axis depth, normals, confidence, semantics, and instance assignments. These predictions connect the image features to 3D surfaces and their owning components. Trajectory-level identity slots separate persistent identity, dimensions, and human shape from frame-dependent root poses and joint rotations. Architectural details and state parameterization are given in Appendix D of the supplementary material.

3.3. Canonical feature aggregation

Spatial supports. A query stores canonical position $\mathbf{q}_i$, cell scale $\boldsymbol{\delta}_i$, type k_i, local attention group g_i, and reference-world coordinate $\mathbf{q}_i^w$. Reliable depth samples provide surface anchors. Background anchors are combined with a finite world grid; rigid anchors are aligned by inverse SE(3) transforms and combined with a grid in the local bounding volume. Human queries lie on the canonical template surface. Sky queries are unit directions grouped by cubemap face. These supports provide explicit locations for both observed and missing content.

The layout remains fixed during completion. We obtain $\mathbf{q}_i^w$ by placing the query in the first input configuration, including skinning for humans. Canonical coordinates identify the content within its component; reference-world coordinates let the generator relate that component to its surroundings.

Motion-aware evidence aggregation. For each observation, we place the canonical query at the corresponding actor state:

$$\mathbf{p}_i(t) = \begin{cases} \mathbf{q}_i, & k_i = \text{background}, \\ \text{xyz}(\mathbf{T}_a^w(t)\bar{\mathbf{q}}_i), & k_i = \text{rigid}, \\ \text{xyz}(\mathbf{T}_h^w(t)\mathbf{A}_i(t)\bar{\mathbf{q}}_i), & k_i = \text{human}. \end{cases} \tag{3}$$

Here $\bar{\mathbf{q}}_i$ is homogeneous, xyz extracts Euclidean coordinates, and $\mathbf{A}_i(t)$ is the weighted skinning transform. Projection gives query depth $z_{i,t,v}$ and samples image features

$\phi_{i,t,v}$, depth $d_{i,t,v}$, and confidence $c_{i,t,v}$. We weight observations by confidence and normalized depth residual $r = (d - z)/(\eta_{\text{abs}} + \eta_{\text{rel}} d)$. The indicator $a_{i,t,v}$ selects visible, depth-consistent observations of the same instance (Appendix D):

$$w_{i,t,v} = a_{i,t,v} c_{i,t,v} \exp(-r_{i,t,v}^2/2), \tag{4}$$

$$\mathbf{h}_i = \frac{\sum_{t,v} w_{i,t,v} [\boldsymbol{\phi}_{i,t,v}; r_{i,t,v}]}{\max(\varepsilon, \sum_{t,v} w_{i,t,v})}. \tag{5}$$

The masks m_i, f_i, and u_i distinguish observed surfaces, known free space, and unknown content, with $m_i + f_i + u_i = 1$. Confidence is the maximum over valid observations; free space is inferred from reliable depth rays. Sky features are aggregated by viewing direction using camera rotation and sky semantics. Appendix D gives the visibility and free-space criteria.

3.4. Latent representation encoding

Type-specific encoders compress the aggregated feature and its spatial support, $[\mathbf{h}_i; \mathbf{q}_i; \boldsymbol{\delta}_i]$, into a latent code $\mathbf{z}_i \in \mathbb{R}^D$. The shared latent dimension lets the completion model exchange information across background, rigid, and human representations. Each query keeps its type, ownership, and coordinate frame even though all codes have the same dimension. Visibility masks and confidence are supplied separately to the completion model. Partial and complete observations therefore use the same intrinsic encoding map. Before conditioning, unobserved codes are set to zero; the observed subset is denoted $\mathbf{z}^{\text{obs}}$.

3.5. Joint latent-space completion

Generative completion module. Our generative completion module (GCM) combines a scene-conditioned flow transformer with constrained sampling (Fig. 4). We parameterize its completion velocity with a DiT-style transformer [26] trained by flow matching [18]. Its input combines current and observed codes, masks, confidence, type, both coordinate descriptions, cell scales, and pooled log features. Background groups follow spatial neighborhoods, each asset forms a local group, and sky directions are grouped by cubemap face. Local self-attention processes these groups separately. A cross-component memory (CCM) module then exchanges context through shared memory:

$$\mathbf{M}' = \mathbf{M} + \text{Attn}_r(\mathbf{M}, \widetilde{\mathbf{X}}, \widetilde{\mathbf{X}}), \tag{6}$$

$$\mathbf{U} = \widetilde{\mathbf{X}} + \text{Attn}_w(\text{LN}(\widetilde{\mathbf{X}}), \mathbf{M}', \mathbf{M}'). \tag{7}$$

Here $\widetilde{\mathbf{X}}$ concatenates locally updated tokens, and attention arguments are queries, keys, and values. A residual feed-forward update follows $\mathbf{U}$. Flow-time modulation enters the local normalization; learned tokens and log summaries initialize memory. This structure keeps each support's local organization while allowing a rigid or human query to use information from the surrounding scene.

Flow training and constrained sampling. Let $\mathbf{z}_1$ be a complete target on the same query layout and $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$ be Gaussian noise. We interpolate only the unknown content at flow time $\tau \in [0, 1]$:

$$\mathbf{z}_\tau = m \odot \mathbf{z}^{\text{obs}} + u \odot [(1 - \tau)\boldsymbol{\epsilon} + \tau \mathbf{z}_1]. \tag{8}$$

Observed codes remain fixed and free-space codes remain zero. The target velocity for unknown queries is $\mathbf{z}_1 - \boldsymbol{\epsilon}$. With an optional loss selector s_i, the objective is

$$\mathcal{L}_{\text{flow}} = \frac{\sum_i u_i s_i \|\mathbf{v}_\theta(\mathbf{z}_\tau, \tau, C)_i - (\mathbf{z}_{1,i} - \boldsymbol{\epsilon}_i)\|_2^2}{D \max(1, \sum_i u_i s_i)}. \tag{9}$$

Complete scene or asset examples supply $\mathbf{z}_1$ at the unknown queries, while s_i selects their contribution to the training loss (Sec. 3.8).

At inference, we initialize unknown codes with noise and take S Euler steps with $\Delta\tau = 1/S$:

$$\mathbf{z}^{j+1} = \Pi_{m,f} \left[\mathbf{z}^j + \Delta\tau \, u \odot \mathbf{v}_\theta(\mathbf{z}^j, j/S, C)\right]. \tag{10}$$

The projection $\Pi_{m,f}$ restores observed codes and clears free-space codes after every step. The seed fixes one sampled completion. Sampling precedes decoding, and the resulting scene is reused when the camera or actor states change.

3.6. Type-specific Gaussian decoding

Background, rigid, and human Gaussian decoders independently map each query to K child Gaussians confined to its support cell. Human children inherit the query's skinning weights; rigid children share their asset's rigid transform. Opacity is multiplied by $1 - f_i$, so free queries cannot produce visible Gaussians. A separate sky head predicts cubemap colors within the observation-supported domain V_{sky}. Gaussian and appearance parameterizations are detailed in Appendix E of the supplementary material.

Each query is decoded independently. Freezing the codec and retaining observed codes therefore preserves the Gaussian parameters of observed surfaces.

3.7. Scene composition

State-driven rendering. At time step k, state s_k specifies the ego pose, active assets, rigid transforms, and human root and joint poses. Rigid poses transform canonical Gaussian centers and covariances rigidly. Human joint poses deform

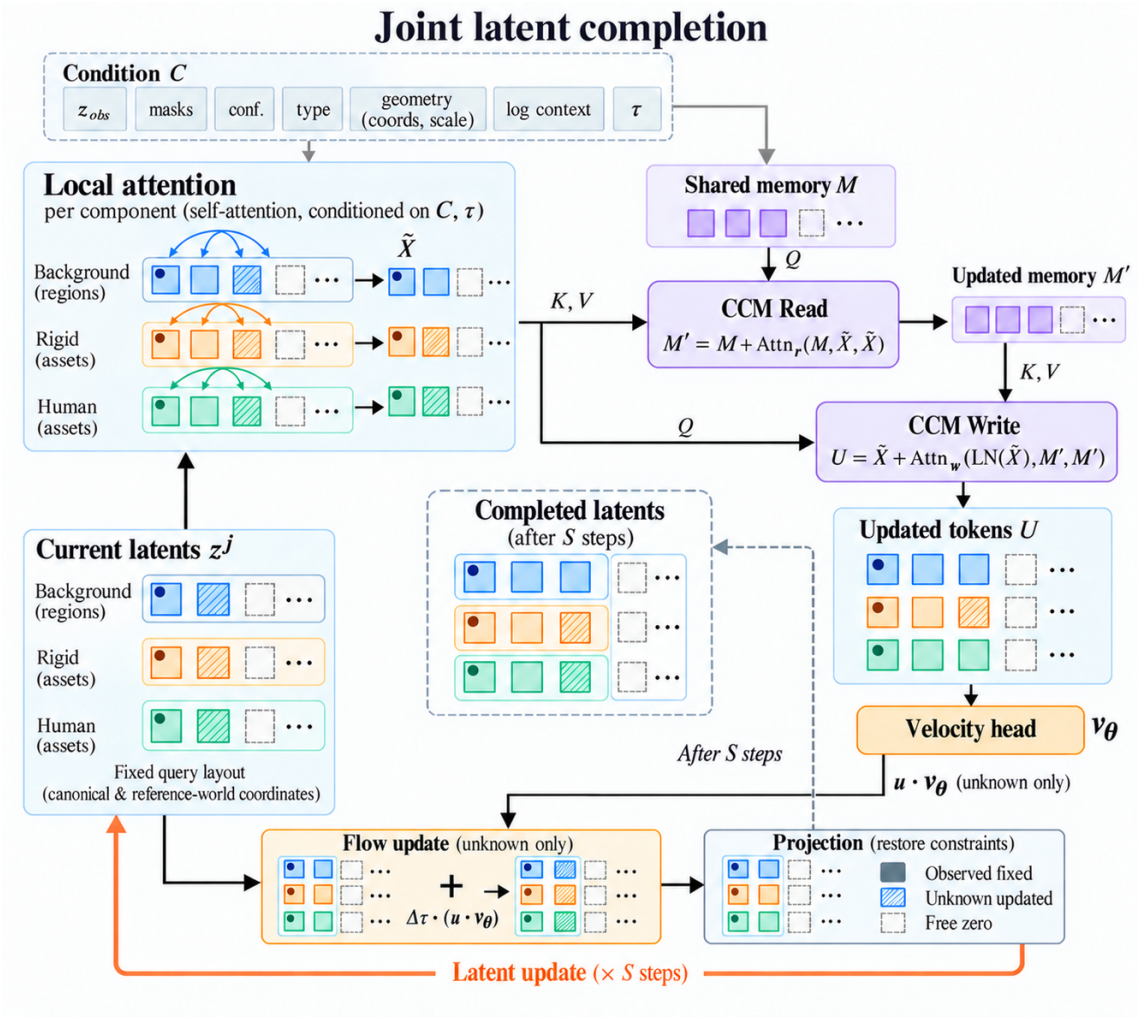

Figure 4. **Joint latent completion.** Group-local attention updates component features; CCM reads them into shared memory and writes context back to every component. The velocity head drives unknown-code updates. Projection restores observed codes and clears free-space codes after each step, with the query layout fixed. Completed codes are taken after the final projection.

them through the stored skinning weights, so completed surfaces move with the corresponding body parts. Appendix C gives the transformations. The renderer composites the background, active assets, and valid sky:

$$o_k = \mathcal{R}(\mathrm{Compose}(\mathcal{W}, s_k), \mathcal{C}_k). \tag{11}$$

Changing the camera selects a new view of the composed scene; changing an actor state moves its completed canonical asset. The background and canonical content remain fixed across these operations, allowing the same reconstruction to be reused along different trajectories.

3.8. Training procedure

We train visual recovery, the latent codec, and the completion model in sequence, then adapt completion through rendered observations (Table 1). Freezing the codec during completion training keeps its target latent space fixed.

Table 1. Training schedule. R: recovery; C: codec; G: GCM. T: trained; F: frozen; –: unused.

Stage	R	C	G	Supervision
Recovery	T	–	–	Geometry, semantics, identity, states
Codec	F	T	–	Partial / complete observations
Prior	F	F	T	Complete targets
Adaptation	F	F	T	Real-log rendering and scene constraints

For codec fitting, partial input and more complete ob-

servations share the same query positions, types, order, and ownership. With recovery and the entire codec frozen, complete scene or asset examples provide a valid target for every unknown query to train the GCM with Eq. 9. Real-log adaptation differentiates through flow integration, decoding, and rendering to update only the GCM. Complete-target samples can be interleaved to retain the prior, with each loss applied only to samples carrying its required targets. Observed anchors are enforced by frozen decoding and hard code restoration. Detailed supervision, losses, and conditioning perturbations are provided in Appendix B of the supplementary material.

4. Experiments

We evaluate our method along three axes: *(i)* dynamic scene reconstruction quality on Waymo Open Dataset (§ 4.2); *(ii)* off-trajectory view synthesis under lateral camera shifts (§ 4.3); and *(iii)* ablations of the main design choices (§ 4.4). Additional cross-dataset generalization results are provided in the supplementary material.

4.1. Experimental Setup

Datasets and protocol. We use the Waymo Open Dataset with 798 training and 202 validation sequences. Following STORM, each validation sequence is divided into non-overlapping 2-second clips of 20 frames at 10 Hz [45]. The FRONT, FRONT_LEFT, and FRONT_RIGHT cameras provide observations at zero-based frame indices $\{0, 5, 10, 15\}$. The remaining 16 frames are used for evaluation: 12 frames for temporal interpolation and four frames for prediction beyond the last observation. Predictions and reference images are resized to 240×160 pixels for evaluation [14].

Baselines. We compare with STORM, DGGT, StreetForward [3, 45, 48]. Methods use the same context timestamps and target views, with model-specific preprocessing. Camera parameters, additional geometric inputs, and rendering adapters are documented with the implementation settings. For spatial extrapolation, we compare Gaussian renderings before diffusion refinement, following PointForward [6].

Metrics. We report PSNR and SSIM over the full image and shared annotated regions. Depth quality is measured using AbsRel and δ_1 at valid projected LiDAR samples after per-frame scale alignment [14]. Scores are averaged over frames within each clip and then over clips within each sequence; sequences receive equal weight. Novel-trajectory renderings are evaluated using FID and FVD against a fixed real-image and video reference population.

Implementation details. DriveCompose reconstructs category-specific actor assets from motion-aligned observations and composes them with a static background. Reconstruction time includes preprocessing, asset generation, and scene composition. Rendering time is measured separately, and asset reuse is evaluated across repeated scene queries.

4.2. Dynamic Scene Reconstruction

We evaluate reconstruction at unobserved times along the recorded camera trajectory. Table 2 reports full-image, dynamic-region, and depth quality; Fig. 5 compares rendered appearance across five examples.

Quantitative comparison. DriveCompose achieves the best scores across all six metrics in Table 2. Full-image and dynamic-region PSNR exceed the strongest baseline in each category by 1.154 and 2.340 dB, respectively. The corresponding SSIM scores reach 0.901 and 0.862, while depth AbsRel decreases to 0.095 and δ_1 reaches 0.877.

Qualitative comparison. Figure 5 shows that DriveCompose retains the pedestrian silhouette in the first row and the rear profile of the central vehicle in the second row, while preserving the surrounding street layout. The baseline renderings exhibit displaced or repeated actor contours in the upper rows and blurred or distorted road structure in the lower rows. DriveCompose recovers these reference structures more closely. Residual artifacts occur around the pedestrians in the third row and the parked pickup in the last row.

Table 2. Dynamic scene reconstruction on Waymo. Image quality is reported over the full image and annotated regions; depth is evaluated at valid LiDAR samples.

	Full image		Dynamic Region		Depth	
Method	PSNR ↑	SSIM ↑	PSNR ↑	SSIM ↑	AbsRel ↓	δ_1 ↑
STORM	26.380	0.826	22.100	0.624	0.178	0.741
DGGT	27.410	0.794	22.710	0.651	0.249	0.668
StreetForward	27.010	0.818	20.919	0.592	0.197	0.759
Ours	**28.564**	**0.901**	**25.050**	**0.862**	**0.095**	**0.877**

4.3. Off-Trajectory View Synthesis

We evaluate off-trajectory rendering under lateral shifts of ± 1, ± 2, and ± 4 meters, keeping camera orientation, intrinsics, and timestamps fixed [6, 51].

Table 3 reports FID and FVD averaged over separately evaluated left and right shifts at each magnitude, using the same real reference population across methods. Figure 6 presents qualitative view-extrapolation results on Waymo

Figure 5. **Qualitative comparison for dynamic scene reconstruction.** Each row shows one example. Columns, from left to right, are ground truth, DGGT, STORM, StreetForward, and DriveCompose (Ours). The examples cover pedestrians, traffic, and urban backgrounds, allowing comparison of actor silhouettes, vehicle boundaries, and road and building structure.

Table 3. Novel-view synthesis under lateral camera shifts. FID and FVD are averaged over left and right shifts at each displacement magnitude. Lower is better.

Method	Shift ±1 m		Shift ±2 m		Shift ±4 m	
	FID ↓	FVD ↓	FID ↓	FVD ↓	FID ↓	FVD ↓
STORM	61.97	722.27	83.72	1029.36	132.57	1704.16
DGGT	73.99	984.00	93.36	1197.03	128.01	1661.70
StreetForward	84.30	878.95	103.31	1095.02	123.58	1578.63
Ours	**28.52**	**544.27**	**37.43**	**673.17**	**58.19**	**817.13**

with the camera shifted 2 m to the left, illustrating vehicle shape, road layout, and occlusion boundaries across three scenes.

4.4. Ablation studies

Protocol. We evaluate both ablations on held-out Waymo images using a shared 29-timestamp window. Each window uses three synchronized cameras: front, front-left, and front-right. The nested input sets are $\mathcal{T}_2 = \{0, 28\}$, $\mathcal{T}_4 = \{0, 8, 20, 28\}$, and $\mathcal{T}_8 = \{0, 4, \ldots, 28\}$, with shared target timestamps $\{2, 6, \ldots, 26\}$. Target images are excluded from evidence and query construction; target camera and actor states are supplied equally to every variant. The recovery module and codec are frozen, with matched query layouts, data exposure, and optimization budgets at each input setting. Training samples $T \in \{2, 4, 8\}$ uniformly; completion uses 24 Euler steps. The two learned variants use training seeds $\{0, 1, 2\}$ and world seeds $\{7, 17, 27\}$; the variant w/o GCM is deterministic. Hyperparameters are selected on an internal development split before evaluation on scene- and identity-disjoint data.

Figure 6. **Qualitative view extrapolation on Waymo with a 2 m leftward camera shift.** Each row shows one scene. Columns, from left to right, show the reference image (GT), STORM, DGGT, StreetForward, and DriveCompose (Ours).

Metrics. We report PSNR, SSIM, and LPIPS over the full held-out target images, with PSNR and SSIM additionally evaluated on dynamic regions for component ablations. All variants use the same image preprocessing, evaluation resolution, and annotation-based dynamic-region masks as the main reconstruction experiment. Metrics are computed on all renderable scenes before any output filtering; non-renderable scenes are counted separately. Scores are averaged over frames within each window, then over windows within each sequence, and finally across sequences and training seeds with equal weight.

Completion and scene context. Table 4 compares the full model with two controls. The variant w/o GCM bypasses the generative completion module and disables the opacity of every non-observed, non-sky query. The variant w/o CCM replaces the cross-component memory module with isolated per-component memories at every layer, while sharing parameters and retaining the same global log context. The learned control is retrained through Stages 3–4; memory-related compute differences are recorded. Ours achieves the best scores on all five metrics. Compared with w/o GCM, it raises full-image and dynamic-region PSNR by 1.776 and 2.186 dB, respectively. Relative to w/o CCM, the corresponding gains are 0.627 and 0.934 dB, while full-image LPIPS decreases from 0.147 to 0.129.

Input sparsity. Table 5 reuses checkpoints of the full model and the variant w/o CCM at $T = 2, 4, 8$, with 6/12/24 input images. The time span and target views remain fixed; evidence and queries are rebuilt from each input subset. No model is retrained for this comparison. Both variants improve across all three metrics as T increases. Ours outperforms w/o CCM at every input setting, with PSNR gains of 1.104, 0.738, and 0.627 dB at $T = 2, 4, 8$, respectively. The LPIPS reduction also increases as inputs become sparser, from 0.018 at $T = 8$ to 0.041 at $T = 2$.

Table 4. **Component ablations.** All variants use $T = 8$ (24 images), shared recovery and codec, and the same target views and dynamic-region masks. Full-image quality is evaluated with PSNR, SSIM, and LPIPS; dynamic-region quality with PSNR and SSIM.

Variant	Full image			Dynamic regions	
	PSNR ↑	SSIM ↑	LPIPS ↓	PSNR ↑	SSIM ↑
w/o GCM	27.137	0.867	0.183	23.431	0.811
w/o CCM	28.286	0.893	0.147	24.683	0.849
Ours	**28.913**	**0.911**	**0.129**	**25.617**	**0.873**

4.5. State-driven reuse

At timestamp 14 and world seed 7, the interventions independently translate the camera by 1 m along its local $+x$ axis, rotate a vehicle by 30° about its root and the world vertical axis, or apply a held-out valid human pose H_1 with fixed root placement. Lighting and canonical assets remain fixed, without resampling completion.

Table 5. **Input-timestamp sensitivity.** At each T, we compare w/o CCM with the full model using fixed checkpoints, time span, and target views. All metrics are evaluated on full images. The $T = 8$ setting is shared with Table 4.

T	Variant	PSNR ↑	SSIM ↑	LPIPS ↓
2	w/o CCM	25.237	0.817	0.254
	Ours	**26.341**	**0.851**	**0.213**
4	w/o CCM	27.126	0.869	0.187
	Ours	**27.864**	**0.891**	**0.159**
8	w/o CCM	28.286	0.893	0.147
	Ours	**28.913**	**0.911**	**0.129**

5. Discussion

DriveCompose separates canonical scene content from camera and actor states. Its representation covers a finite region under fixed illumination, with rigid vehicles and humans bound to a fixed skeleton and clothing volume. Depth, identity, and canonical-pose estimates determine the observed anchors used by completion, so recovery errors propagate into the reconstructed geometry. Human articulation follows the template's geometric support, and contact and collision behavior are handled by the physical simulator. Relighting and changes in clothing geometry require corresponding appearance and deformation models.

6. Conclusion

We presented DriveCompose, which reconstructs compositional driving scenes through canonical feature aggregation, latent encoding, and joint completion. Local attention and cross-component memory combine component structure with scene context. Type-specific Gaussian decoders retain identity and motion bindings, allowing completed assets to be composed under changing camera and actor states. On Waymo, DriveCompose improves dynamic reconstruction and off-trajectory view synthesis.

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